MODEL QUESTION PAPER

REVISION-2010 Reg. No…………………

SUB CODE: 3076 Signature……………….

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING / TECHNOLOGY

INSTRUMENT TRANSDUCERS

(Common for EI & IT)

Time : 3 hours Max. marks: 100

PART – A

(Answer all questions in one or two sentences. Each question carries 2 marks)

Marks

I

1. Differentiate sensor and transducer.
2. Define gauge factor.
3. What is Villari effect?
4. List four materials that exhibit peizo-electric effect.
5. What is photo-electric effect? (5 x 2 = 10)

PART – B

(Answer any five questions. Each question carries 6 marks)

II

1. List the advantages of electric transducers.
2. Explain loading effect with an example.
3. Define residual voltage and its causes.
4. Explain the working principle of Hall-effect transducer.
5. Describe how level of non-conducting liquid can be measured using capacitive transducer.
6. Draw the equivalent circuit of a peizo- electric transducer and define charge sensitivity.
7. Sketch the characteristic of an ionization chamber and explain the regions.

(5 x 6 = 30)

PART – C

(Answer one full question from each unit. Each question carries 15 marks.)

UNIT – I

III

1. Explain the classification of transducers. 10
2. A resistance wire strain gauge with a gauge factor of 2 is bonded to a steel structural member subjected to a stress of 100 MN/m^2. The modulus of elasticity of steel is 200 GN/m^2. Calculate the percentage change in the value of the gauge resistance due to the applied stress. 5

OR

IV	Marks
1. Derive the expression for loading error for a linear potentiometer.	9
2. Explain how strain can be measured using Wheatstone's bridge circuit	6

UNIT – II

V	
1. Explain the construction and working of LVDT.	8
2. Describe the applications of Hall effect transducer.	7

OR

VI	
1. Explain the working of variable reluctance transducer.	8
2. Outline the principle of operation of Magneto resistive transducer.	7

UNIT – III

VII	
1. Describe the differential arrangement used in capacitive transducer.	9
2. Explain the working of peizo-electric accelerometer.	6

OR

VIII	
1. A capacitive transducer uses two quartz diaphragms of area $750 mm^2$ separated by a distance of 3.5 mm. A pressure when applied to the top diaphragm produces a deflection of 0.6 mm. The capacitance is 370 pF when no pressure is applied to the diaphragms. Find the value of capacitance after the application of pressure.	7
2. Derive the expression for output voltage of a peizo-electric transducer.	8

UNIT – IV

IX	
1. Describe the working principle of photo-emissive cell.	7
2. Explain the construction and operation of Geiger-Muller counter.	8

OR

X	
1. Describe the working principle of photomultiplier tube.	7
2. Explain the construction and operation of proportional counter.	8

(4 x 15 = 60)